

Relatório ELM

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1 Introduction

Extreme Learning Machines (ELMs) are single-hidden layer feedforward neural networks where input weights are randomly initialized, and output weights are analytically determined. This report explores the impact of regularization on ELM's decision boundaries using synthetic datasets.

2 Methodology

The ELM classifier implemented includes:

- Random initialization of input weights and biases with Glorot scaling.
- Activation functions: Sigmoid, ReLU, Tanh, and RBF.
- Regularized least squares for output weights: $\beta = (\mathbf{H}^T \mathbf{H} + \lambda \mathbf{I})^{-1} \mathbf{H}^T \mathbf{Y}$, where λ is the regularization parameter.

Two experiments were conducted:

1. A synthetic 2D dataset using `make_classification` with an 80-20 train-test split.
2. The moons dataset with varying regularization parameters to visualize decision boundaries.

3 Results

3.1 Synthetic Dataset

The ELM achieved a test accuracy of **94.00%** on the synthetic dataset, demonstrating effective separation of classes.

3.2 Decision Boundaries with Regularization

Figure 1 shows decision boundaries for the moons dataset under different regularization strengths:

Key observations:

- $\lambda = 0$: Complex boundary, potential overfitting.
- $\lambda = 0.001$: Smoother boundary, improved generalization.

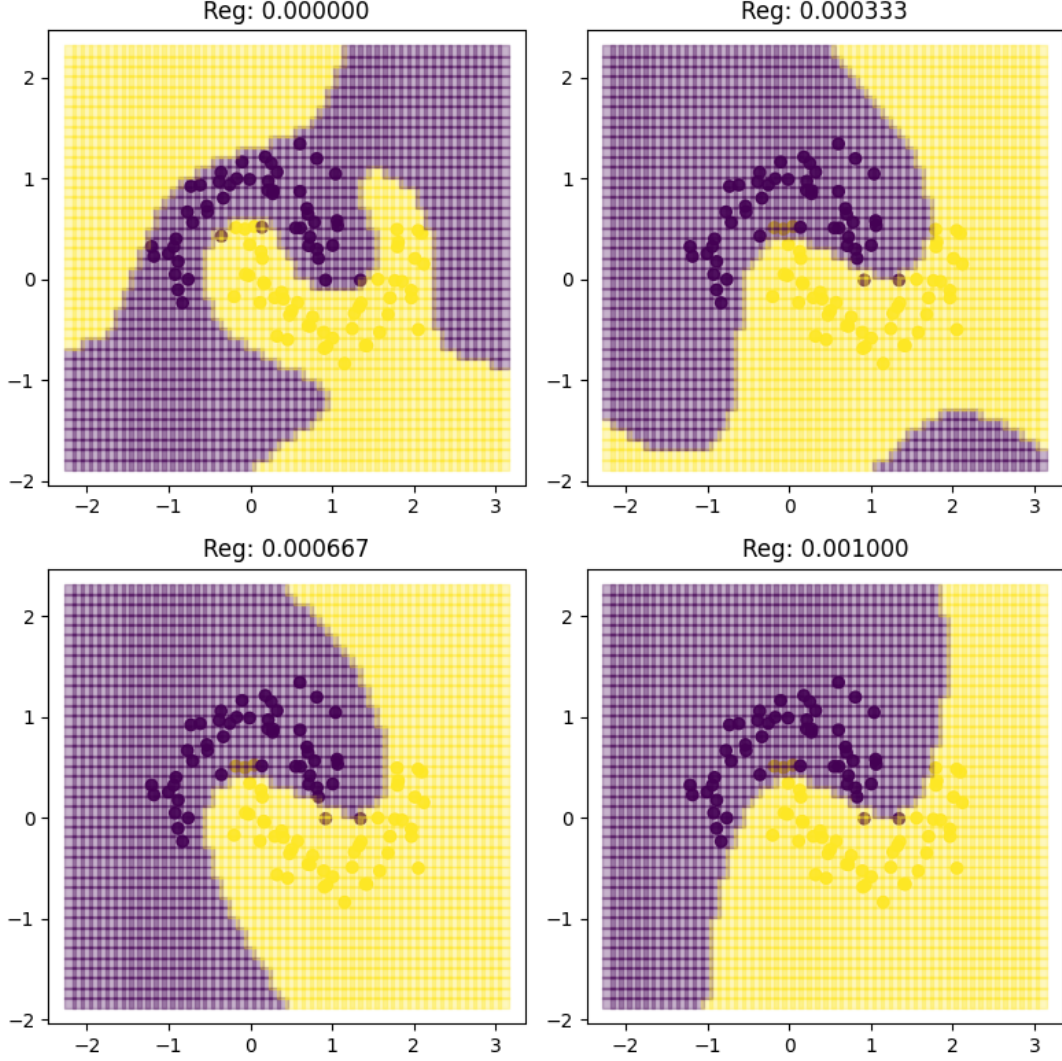


Figure 1: Decision boundaries with varying regularization (λ). Top-left to bottom-right: $\lambda = 0, 0.000333, 0.000667, 0.001$. Higher λ smooths boundaries, reducing overfitting.

4 Conclusion

Regularization in ELM controls model complexity by penalizing large weights. Higher λ values yield smoother decision boundaries, trading training accuracy for better generalization. The experiments validate the importance of tuning λ to balance bias and variance.